Metadata and Provenance Analysis of an Image File

Case study: `Mostrando_los_productos_de_la_milpa.jpg` · Prepared by: Mohammad Hanif Alghifari · 18 October 2026

1. Summary

This report traces the origin and integrity of a single JPEG file (4592 × 3056 pixels, 5,582,789 bytes) hosted on Wikimedia Commons. The approach: read the EXIF and MakerNotes metadata, test internal consistency with simple calculations, compare the file against the Commons page and the related Flickr profile, and then separate clearly what is **proven**, what is a **hypothesis**, and what is **unknown**.



Figure 1. The analyzed photo (a 1714 × 2576 pixel preview copy of the uploaded file, rotated according to the Orientation tag). The original file whose metadata was analyzed measures 4592 × 3056 pixels (5,582,789 bytes); this preview copy is not byte-for-byte identical to it.

Key findings:

1. The metadata is internally consistent with a Sony DSLR-A350 camera and a Minolta/Sony AF DT 18-70mm F3.5-5.6 (D) lens.
2. No trace of editing software was found in the metadata. This is not conclusive proof that the file is unaltered, since metadata can be stripped or forged.
3. Provenance is incomplete: the photographer is unknown, and the date of 12 October 2013, 18:38:39 comes only from the camera clock, with no time zone and no independent confirmation.
4. This report includes a corrections section (Section 11) for several conclusions in the first draft that turned out to be overstated. The corrections are kept on purpose because they are part of the method.

2. Scope and Ethics

What is analyzed is the **file**, not a person. The photo contains a recognizable face. The Commons caption (“Campesino de Michoacán”) is only a description, not an identity, and tracing the subject’s identity was **deliberately not attempted**. Tracing an individual from a photo is doxing, not provenance analysis. For real engagements, written permission, an agreed scope, and compliance with applicable data protection law are required.

3. Materials and Method

Materials: the image file; a metadata report from ExifTools.com (PDF, generated 18 Oct 2026 at 07:16 UTC); three screenshots (the About and Photostream pages of the Flickr account “Feria de Productores”, and the file page on Wikimedia Commons).

Method: (a) reading the metadata; (b) cross-tag internal consistency tests; (c) cross-checking against Commons and Flickr; (d) an astronomical plausibility test of the time (manual estimate); (e) tiered classification of conclusions.

Limitation: the analysis is based on the PDF report and screenshots, not on running ExifTool locally. Hash values come from ExifTools.com.

4. File Identity

Property	Value
Type	image/jpeg (JPEG, old-style compression)
Size	5,582,789 bytes (5.58 MB base 10 = 5.32 MiB base 2)
Dimensions	4592 × 3056 (14 MP, 3:2 ratio), Orientation: Rotate 90 CW
MD5	`e40bc449f8b3e9047acacf74215e6820`
SHA-1	`16dafb90a789733265f907ba8b10b4bb82977ccd`
SHA-256	`e157d24196a67d2e183b05218e875f8428afdee6dc3b6f8d7af7cd566fad5ebe`

A hash is a fingerprint of every byte in the file; anyone holding an identical file can match it to this report.

5. Device and Consistency Tests

Camera: SONY DSLR-A350, firmware string `DSLR-A350 v1.00`. Lens: Minolta/Sony AF DT 18-70mm F3.5-5.6 (D). Quality Fine, size Large.

Test	Calculation	Result
35 mm equivalent focal length	18.0 mm × 1.5	27 mm, matches
Maximum aperture	MaxApertureValue 3.5 at 18 mm	matches lens specification
Resolution	$4592 \times 3056 \div 10^6$	14.0 MP, matches
Light value	$\log_2(6.3^2 \div (1/50))$	about 10.96, matches LightValue 11
Dimensions	ExifImageWidth/Height vs ImageSize	identical

All tests pass. There is no sign that this file is a composite of several sources.

6. Shooting Conditions

Parameter	Value
Mode	Program AE, SceneMode Auto
Aperture / Shutter / ISO	f/6.3 · 1/50 s · ISO 100
Exposure compensation	0 EV
Metering	Multi-segment
Flash	Off, did not fire
Focus	AF-A, FocusStatus Confirmed
Focus distance	0.342 m; DOF 0.09 m (0.30–0.39 m)
Stabilization	On

The subject is very close and the in-focus zone is only about 9 cm deep. This agrees with how the photo looks: the hand and pods are sharp while the field behind melts away.

7. Integrity Tests

Indicators supporting camera origin:

- `FileSource: Digital Camera` and `SceneType: Directly photographed`.
- `Software` holds a firmware string, not the name of an editing application.
- `CreateDate`, `ModifyDate`, and `DateTimeOriginal` are identical to the second.
- `UserComment` is empty; `InteropIndex` is R98 (DCF, sRGB).
- The ThumbnailImage and PreviewImage embedded in the report appear to share the main image's composition (rotated per Orientation). This is a rough visual check, not a pixel-level analysis.

Limitation: every indicator above is metadata, and metadata can be deleted, rewritten, or forged. The defensible conclusion is *no trace of editing in the metadata*, not *proven unedited*.

Authentic is not neutral. The camera had already processed the pixels before the file left the device: Vivid style, Dynamic Range Optimizer Advanced Auto, contrast +1, saturation +2, sharpness +1. Colors in the photo must not be used as evidence of field conditions.

8. Anomalies and How to Interpret Them

8.1 EXIF vs MakerNotes

Parameter	EXIF	MakerNotes
Aperture	f/6.3	f/6.7
Shutter	1/50	1/54
Shutter setting	–	1/256
Aperture setting	–	5.7

The roughly one-third-stop difference in aperture and shutter is *probably* due to different rounding or quantization between blocks (hypothesis). The `ShutterSpeedSetting` of 1/256 and `ApertureSetting` of 5.7 differ greatly, and their meaning in Program AE mode cannot be determined from this data. Working rule: never take one column and treat it as final. A `Brightness` of 0 in MakerNotes is likely a "not available" marker, not "pitch dark".

8.2 LensSpec not populated

LensType holds the lens name, but LensSpec reads Unknown (00 0 0 0 0 00). This is probably a firmware quirk. There is no evidence of manipulation.

8.3 Settings that do not apply at this moment

HighISONoiseReduction High (although ISO is 100), LongExposureNoiseReduction On (although the shutter was 1/50 s), RedEyeReduction On (although the flash was off), and ColorTemperatureSet 2500 K (although white balance was Auto). The lesson: **settings prove what the owner switched on, not what actually happened.**

8.4 Autofocus status data

Of the eleven sensors reported, nine show Front Focus (from −355 at Top-left to −83 at Bottom-left), one shows Back Focus (Bottom, +155), and one shows Out of Focus (Bottom-right). The active sensor (Bottom-left, −83) matches the AFPoint value.

In the first draft this pattern was read as lens misalignment and a “device fingerprint”. **That interpretation is withdrawn** (see Section 11): the values reflect different distances from each sensor to objects within *a single frame* (leaves in front, the subject, the field behind), so they depend on the scene, not on equipment calibration. The data only proves which sensor was active and that focus was confirmed.

9. Provenance

Stage	Evidence	Confidence
Device	Sony DSLR-A350 (introduced 2008) + DT 18-70mm lens	High
Capture	12 Oct 2013, 18:38:39 per the camera clock	Low: no time zone, unverified
Public archive	Commons carries the caption “Campesino de Michoacán mostrando los productos de la milpa (Tangamandapio, Michoacán)” and the date 12 October 2013, 18:38:39	Medium for existence; the caption is a human claim
File size	Commons: 5.32 MB (actually MiB); ExifTools: 5.6 MB	Matches after unit conversion
Flickr account	“Feria de Productores”, joined January 2020, 587 photos, occupation Mercado Local, Guadalajara/Zapopan, Jalisco	High for the profile; link to this file not proven

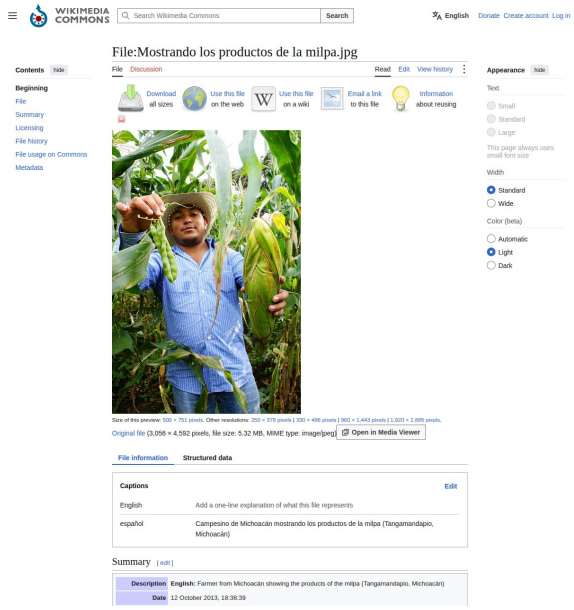


Figure 2. The file page on Wikimedia Commons: Spanish caption, date 12 October 2013 18:38:39, original size 3,056 × 4,592 pixels and 5.32 MB.

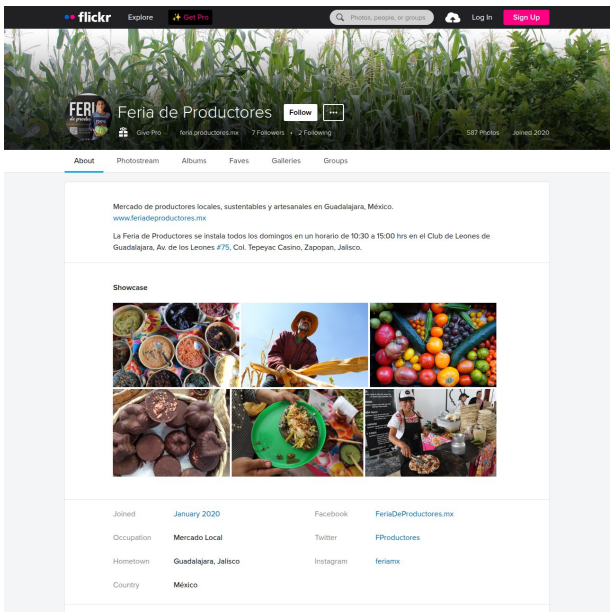


Figure 3. The About page of the Flickr account "Feria de Productores": joined January 2020, 587 photos, Guadalajara, Jalisco.

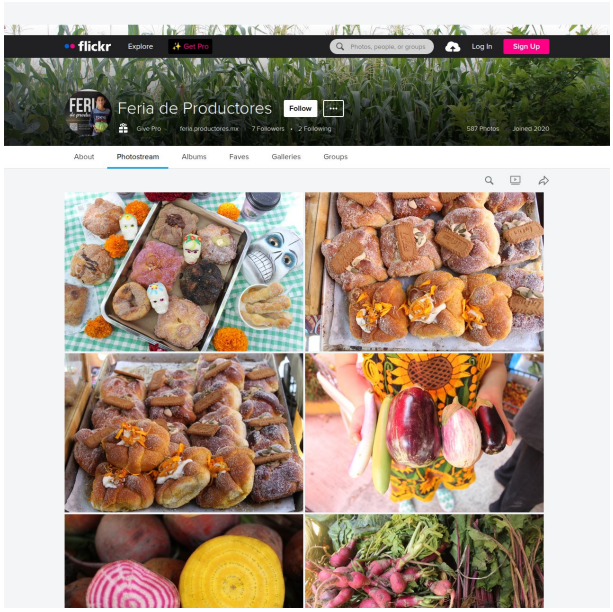


Figure 4. The photostream of the same Flickr account. The analyzed photo is not visible in this screenshot.

Important notes:

- **The Commons date is not independent confirmation.** Because it matches the EXIF to the second, Commons most likely took it from the same EXIF (hypothesis).
- **The photo is not visible** in the screenshots of the Flickr About page or Photostream. The Flickr-to-Commons link cannot be verified from the available evidence; the Source field and file history on Commons are needed.
- An account created in January 2020 only sets the earliest upload date *by that account*. It could still upload an old photo, so the account's age does not determine who the photographer is.
- The Flickr profile mentions Jalisco, while the Commons caption mentions Michoacán. This difference is worth noting, but it is not necessarily a contradiction: the profile describes the market's location, not where every photo was taken. The explanation that the photo was made for a market supplier is **speculation**.

Photographer: not identified. That is a legitimate answer.

A methodological note: a Flickr account's identity is determined by the username in the URL (`feriamx`), not the display name, which anyone can change.

10. Time Plausibility Test

Mexico still observed daylight saving time in 2013 (it ended on 27 October 2013), so 12 October 2013 fell at UTC−5 for the central zone, where Michoacán lies. A manual estimate puts sunset near Tangamandapio at about 19:10–19:30 local time, so 18:38 falls roughly 30–50 minutes before sunset. A LightValue of 11 (shaded light) is **plausible** for an afternoon under a corn canopy.

Plausibility, however, is not proof:

- There is no `OffsetTime`, so 18:38 is a naive time without a time zone.
- The camera clock was never network-synchronized; if the battery was ever removed, the clock may have reset to default without notice.
- There is no GPS data. To my knowledge the DSLR-A350 has no GPS module (to be checked against the official specification).

Final status: not refuted, but not proven. The sunset figures are a manual estimate and should be checked with an ephemeris calculator.

11. Corrections to the First Draft

This section is kept so that the reasoning stays open to inspection.

First-draft claim	Problem	Revision
18:38 is past sunset, so LightValue 11 is impossible	Used a 2026 sunset table for a 2013 event without accounting for daylight saving time	Withdrawn; time is plausible (Section 10)
Flickr is not the original source; the account "could not have taken the photo"	A newer account can upload old photos; the photo is also not visible on the account	Only an earliest-upload bound; photographer unknown
Proven never edited	Metadata cannot prove the absence of editing	No trace of editing in the metadata
Nine of eleven sensors Front Focus means a misaligned lens and a device fingerprint	AF status depends on the scene within one frame	Withdrawn as evidence of calibration
File identical across sources based on hash	The Commons hash is not visible in the evidence; only size and dimensions match	Compare SHA-1 with Commons
1/256 and f/5.7 are leftovers of pre-exposure metering	Not supported by evidence	Marked as an uncertain interpretation
The photo was made for a producers' market	No basis	Speculation

12. Tiered Conclusions

Supported by evidence

- Metadata is consistent with a Sony DSLR-A350 and a DT 18-70mm lens.
- No trace of editing software in the metadata.
- The camera applied Vivid style, DRO, and sharpening, so colors are not a raw representation.

- File size and dimensions match the Commons page.

Hypotheses

- Shot in Michoacán (based on the Commons caption).
- The Commons date was derived from the same EXIF.
- The EXIF vs MakerNotes differences are due to rounding.

Unknown

- The photographer’s identity (and the subject’s identity, deliberately not pursued).
- The actual date, time, and time zone of capture.
- The upload relationship between Flickr and Commons.
- The result of a pixel comparison with ThumbnailImage, PreviewImage, and TiffMeteringImage.

13. Limitations

- Based on a PDF report and screenshots, not direct inspection of the file.
- The behavior of 2008 camera firmware cannot be confirmed without Sony documentation.
- The astronomical estimate is manual.
- The embedded artifacts have not been extracted and compared.

14. Next Steps

1. Extract ThumbnailImage, PreviewImage, and TiffMeteringImage and compare them with the main image.
2. Compare the SHA-1 with the value on the Commons file information page.
3. Check the Source field, file history, and upload log on Commons.
4. Search for appearances of the photo before 2020 (reverse image search) to test the 2013 claim.
5. Compute sunset with an official ephemeris calculator for Tangamandapio’s coordinates.
6. Check the official Sony α350 specification (AF points, absence of GPS).

Appendix A. Reproducible Commands

```
exiftool -a -G1 -u file.jpg                       # all blocks incl. MakerNotes
sha256sum file.jpg; sha1sum file.jpg; md5sum file.jpg   # hashes
exiftool -AFStatus* file.jpg                      # AF sensor status
exiftool -b -ThumbnailImage file.jpg > thumb.jpg
exiftool -b -PreviewImage file.jpg > preview.jpg
exiftool -b -TiffMeteringImage file.jpg > metering.tif
```

Appendix B. Glossary

EXIF: standard data embedded in a photo. **MakerNotes**: camera-brand-specific internal notes. **Hash**: a fingerprint of every byte in a file. **DRO**: a camera feature that adjusts dynamic range and alters pixels. **LightValue**: a measure of scene brightness from aperture, shutter, and ISO. **DOF**: the depth of the zone that appears sharp. **Provenance**: the history of a file’s origin and custody. **Naive time**: a timestamp with no time zone information.